```
#=====================
#   import library
#======================'

import numpy as np

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns


#===================='
#   import
data
#===================='

data=pd.read_excel(r"C:\Users\Elbostan\Desktop\Call-Ce
nter.xlsx")

#======================='
#   Data
preprocessing.
#======================='

data.info()

data.isna().sum()

"fil
l null columns"

data[['Satisfaction rating', 'AvgTalkDuration',
'Speed of answer in seconds']] = data[['Satisfaction rating',
'AvgTalkDuration', 'Speed of answer in
seconds']].fillna(0)


describe=data.describe()
plt.figure(figsize=(10,
6))
sns.heatmap(describe,annot=True, cmap="coolwarm",
linewidths=0.5)
plt.show()

#===============================================================&
#039;
#                        Data
Analysis
#==============================================================='

"
              Overall customer satisfaction ?
"

satisfaction_percentage = data['Satisfaction
rating'].value_counts(normalize=True) * 100
plt.pie(satisfaction_percentage.values,
labels=satisfaction_percentage.index, autopct='%1.1f%%',
startangle=90)
plt.title("satisfaction
percentage")
plt.show()

sns.barplot(x=satisfaction_percentage.index,
y=satisfaction_percentage.values)
plt.ylabel('satisfaction
percentage')
plt.show()



"                     Overall calls answered/abandoned
 ?                       "
```

```
sns.countplot(data=data, x='Answered
(Y/N)')
plt.show()


"                            Calls by time
                 "

data['Date'] =
pd.to_datetime(data['Date'])
data['DateTime'] = data.apply(lambda row:
pd.to_datetime(str(row['Date'].date()) + ' ' +
row['Time'].strftime('%H:%M:%S')), axis=1)
data['Hour'] =
data['DateTime'].dt.hour
calls_by_hour = data.groupby('Hour')['Call
Id'].count().reset_index()
calls_by_hour.rename(columns={'Call Id':
'Number of Calls'}, inplace=True)

sns.barplot(data=calls_by_hour,
x='Hour', y='Number of Calls')
plt.title("calles by
hours")
plt.xlabel("hours")
plt.ylabel("calles
answers")
plt.xticks(rotation=45)
plt.show()
```